

1. Linux Introduction

What is Open Source?

- Open source: software and source code available to all
- The Free Software Foundation specifies four freedoms
 - The freedom to run the program for any purpose.
 - The freedom to study and modify the source code
 - The freedom to redistribute the program
 - The freedom to create derivative programs
- Many open-source licenses exist, each with different particulars

Linux Origins

- 1984: The GNU Project and the Free Software Foundation
 - Creates open source version of UNIX utilities
 - Creates the General Public License (GPL)
 - Software license enforcing open source principles
- 1991: Linus Torvalds
 - Creates open source, UNIX-like kernel, released under the GPL
 - Ports some GNU utilities, solicits assistance online
- Today:
 - Linux kernel + GNU utilities = complete, open source, UNIX-like operating system
 - Packaged for targeted audiences as *distributions*



Sachin Singh ✓
@sachin_Singh

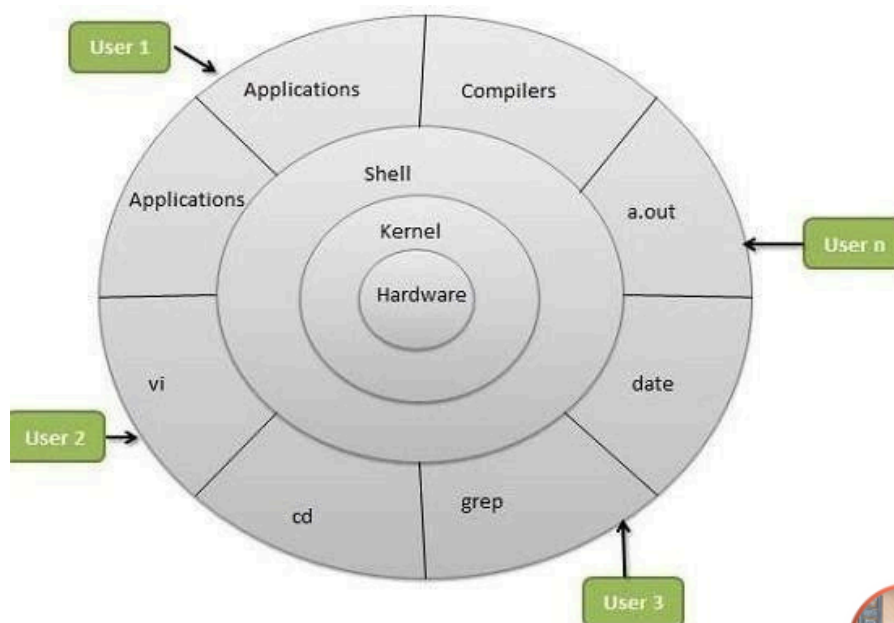
Linux principles

- Everything is a file (including hardware)
- Small, single-purpose programs
- Ability to chain programs together to perform complex tasks
- Avoid captive user interfaces
- Configuration data stored in text

Why Linux?

- OpenSource.
- Community support.
- Heavily customizable.
- Most Servers runs on Linux.
- DevOps most of the tools implements on Linux only.
- Automation
- Secure.

Architecture of Linux



Sachin Singh ✓
@sachin_Singh

Some Important Directories

- Home Directories: `/root`, `/home/username`
- User Executable: `/bin`, `/usr/bin`, `/usr/local/bin`
- System Executables: `/sbin`, `/usr/sbin`, `/usr/local/sbin`
- Other Mountpoints: `/media`, `/mnt`
- Configuration: `/etc`
- Temporary Files: `/tmp`
- Kernels and Bootloader: `/boot`
- Server Data: `/var`, `/srv`
- System Information: `/proc`, `/sys`
- Shared Libraries: `/lib`, `/usr/lib`, `/usr/local/lib`

Diffrent Linux distros

→ Popular Desktop Linux OS

- • Ubuntu Linux
- • Linux Mint
- • Arch Linux
- Fedora
- Debian
- OpenSuse

→ Popular Server Linux OS

- • Red Hat Enterprise Linux
- • Ubuntu Server
- Centos
- SUSE Enterprise Linux



Sachin Singh ✓
@sachin_Singh

Most used Linux distros currently in IT industry.

RPM based:- RHEL & Centos

Debian based :- Ubuntu Server

Difference between RPM based and Debian based.

From user's point of view, there isn't much difference in these tools. The RPM and DEB formats are both just archive files, with some metadata attached to them. They are both equally arcane, have hardcoded install paths and only differ in subtle details. DEB files are installation files for Debian based distributions. RPM files are installation files for Red Hat based distributions. Ubuntu is based on Debian's package management system, APT and DPKG. Red Hat, CentOS and Fedora are based on the old Red Hat Linux package management system, RPM.

DEB or .deb (Debian based softwares)

DEB is the extension of the Debian software package format and the most often used name for such binary packages. DEB was developed by Debian.

Example: Google chrome software

Package name: google-chrome-stable_current_amd64.deb

Installation: `dpkg -i google-chrome-stable_current_amd64.deb`

RPM or .rpm (Red Hat based softwares.)

It is a package management system. The name RPM variously refers to the .rpm file format, files in this format, software packaged in such files, and the package manager itself. RPM was intended primarily for Linux distributions; the file format is the baseline package format of the Linux Standard Base. RPM was developed by Community & Red Hat.

Example: Google chrome software

Package Name: google-chrome-stable-57.0.2987.133-1.x86_64.rpm

Installation: `rpm -ivh google-chrome-stable-57.0.2987.133-1.x86_64.rpm`

NOTE: You will also encounter different commands, packages both kinds of distros.



Sachin Singh ✓
@sachin_Singh

2. Basic Commands

→ Open Terminal

→ Know where you are? Present Working Directory

```
imran@DevOps: ~  
File Edit View Search Terminal Help  
imran@DevOps:~$ pwd  
/home/imran  
imran@DevOps:~$
```

→ Create a directory/folder in your home directory.

```
imran@DevOps:~$ mkdir linux-practices  
imran@DevOps:~$
```

→ Change your current working directory to linux-practices(Go to linux-practices folder).

```
imran@DevOps:~$ cd linux-practices/  
imran@DevOps:~/linux-practices$
```

→ Create some more directories and list them with “ls” command.

```
imran@DevOps:~/linux-practices$ mkdir vmdir  
imran@DevOps:~/linux-practices$ mkdir testdir  
imran@DevOps:~/linux-practices$ mkdir devopsdir  
imran@DevOps:~/linux-practices$ ls  
devopsdir testdir vmdir
```

→ Create some empty files with “touch” command and list them.

```
imran@DevOps:~/linux-practices$ touch file2 file3 file4  
imran@DevOps:~/linux-practices$ ls  
devopsdir file1 file2 file3 file4 testdir vmdir
```

→ Reconfirm your location in your system.

```
imran@DevOps:~/linux-practices$ pwd
/home/imran/linux-practices
imran@DevOps:~/linux-practices$ ls
devopsdir  file1  file2  file3  file4  testdir  vpsdir
```

Absolute path and Relative path

What is a path?

A path is a unique location to a file or a folder in a file system of an OS. A path to a file is a combination of / and alpha-numeric characters.

What is an absolute path?

An absolute path is defined as the specifying the location of a file or directory from the root directory(/). In other words we can say absolute path is a complete path from start of actual filesystem from / directory.

Some examples of absolute path:

/home/imran/linux-practices/

/var/ftp/pub

/etc/samba.smb.conf

/boot/grub/grub.conf

If you see all these paths started from / directory which is a root directory for every Linux/Unix machines.

What is the relative path?

Relative path is defined as path related to the present working directory(pwd). Suppose I am located in /home/imran and I want to change directory to /home/imran/linux-practices. I can use relative path concept to change directory to linux-practices and also devopsdir directory.

```
File Edit View Search Terminal Help
imran@DevOps:~$ pwd
/home/imran
imran@DevOps:~$ cd linux-practices/
imran@DevOps:~/linux-practices$ ls
devopsdir  file1  file2  file3  file4  testdir  vpsdir
imran@DevOps:~/linux-practices$ pwd
/home/imran/linux-practices
imran@DevOps:~/linux-practices$ cd devopsdir/
imran@DevOps:~/.../devopsdir$ pwd
/home/imran/linux-practices/devopsdir
imran@DevOps:~/.../devopsdir$
```


If you see all these paths did not start with / directory.

→ Creating directories in devopsdir directory with absolute and relative path.

```
File Edit View Search Terminal Help
imran@DevOps:~/linux-practices$ ls
devopsdir file1 file2 file3 file4 testdir vpdire
imran@DevOps:~/linux-practices$ mkdir devopsdir/ansible
imran@DevOps:~/linux-practices$ mkdir /home/imran/linux-practices/devopsdir/aws
imran@DevOps:~/linux-practices$ ls devopsdir/
ansible aws
imran@DevOps:~/linux-practices$
```

→ Copying files into directory.

```
File Edit View Search Terminal Help
imran@DevOps:~/linux-practices$ pwd
/home/imran/linux-practices
imran@DevOps:~/linux-practices$ ls
devopsdir file1 file2 file3 file4 testdir vpdire
imran@DevOps:~/linux-practices$ cp file1 testdir/
imran@DevOps:~/linux-practices$ cd testdir/
imran@DevOps:~/.../testdir$ ls
file1
imran@DevOps:~/.../testdir$
```

→ Copying directories from one location to another.

```
File Edit View Search Terminal Help
imran@DevOps:~/linux-practices$ cd
imran@DevOps:~$ cd -
/home/imran/linux-practices
imran@DevOps:~/linux-practices$ pwd
/home/imran/linux-practices
imran@DevOps:~/linux-practices$ ls
devopsdir file1 file2 file3 file4 testdir vpdire
imran@DevOps:~/linux-practices$ cp -rvfp testdir/ vpdire/
'testdir/' -> 'vpdire/testdir'
'testdir/file1' -> 'vpdire/testdir/file1'
imran@DevOps:~/linux-practices$ ls vpdire/
testdir
imran@DevOps:~/linux-practices$
```

→ Moving files from one location to another.

```
imran@DevOps: ~/linux-practices
File Edit View Search Terminal Help
imran@DevOps:~/linux-practices$ pwd
/home/imran/linux-practices
imran@DevOps:~/linux-practices$ ls
devopsdir  file1  file2  file3  file4  testdir  vpdire
imran@DevOps:~/linux-practices$ mv devopsdir/ vpdire/
imran@DevOps:~/linux-practices$ ls
file1  file2  file3  file4  testdir  vpdire
imran@DevOps:~/linux-practices$ ls vpdire/
devopsdir  testdir
imran@DevOps:~/linux-practices$
imran@DevOps:~/linux-practices$ mv file3 file4 vpdire/
imran@DevOps:~/linux-practices$ ls
file1  file2  testdir  vpdire
```

→ Removing files and directories.

```
imran@DevOps:~/linux-practices$ rm file1
imran@DevOps:~/linux-practices$ ls
file2  testdir  vpdire
imran@DevOps:~/linux-practices$ rm -rf testdir/
imran@DevOps:~/linux-practices$ ls
file2  vpdire
```


VIM EDITOR

→ Install vim editor.

```
imran@DevOps:~/linux-practices$ sudo apt-get install vim
[sudo] password for imran:
Reading package lists... Done
Building dependency tree
Reading state information... Done
```

→ Open up a file in vim editor

```
imran@DevOps:~/linux-practices$ vim firstfile.txt
```

→ Hit i to enter into insert mode

```
imran@DevOps: ~/linux-practices
File Edit View Search Terminal Help
~
~
~
~
-- INSERT --                                0,1      All
```

=> type few lines => hit Esc

```
imran@DevOps: ~/linux-practices
File Edit View Search Terminal Help
This is first line in vim editor.
This one's second
So on
and
So forth.
~
~
6,0-1      All
```

=> type :wq

```
imran@DevOps: ~/linux-practices
File Edit View Search Terminal Help
This is first line in vim editor.
This one's second
So on
and
So forth.
~
:wq
```

=> Enter.

→ Read file with cat command.

```
imran@DevOps: ~/linux-practices
File Edit View Search Terminal Help
imran@DevOps:~/linux-practices$ cat firstfile.txt
This is first line in vim editor.
This one's second
So on
and
So forth.
imran@DevOps:~/linux-practices$
```

VIM EDITOR

- VI Visual display editor
- VIM Visual display editor improved

This is command mode editor for files. Other editors in Linux are emacs, gedit
vi editor is most popular

- It has 3 modes:
- 1 Command Mode
 - 2 Insert mode (edit mode)
 - 3 extended command mode

Note: When you open the vim editor, it will be in the command mode by default.

Command Mode:

gg	To go to the beginning of the page To go to end
G	of the page To move the cursor forward, word
w	by word To move the cursor backward, word by
b	word
nw	To move the cursor forward to n words
nb	(SW)
u	To move the cursor backward to n words
{SB}	
U	To undo the previous changes (entire line) To undo last change (word)
Ctrl+R	To redo the changes
YY	To copy a line
nyy	To copy n lines (Syy or 4yy)
p	To paste line below the cursor position
P	To paste line above the cursor position
dw	To delete the word letter by letter {like Backspace}
x	To delete the world letter by letter (like DEL Key)
dd	To delete entire line
ndd	To delete n no. of lines from cursor position{Sdd}
I	To search a word in the file

Extended Mode: (Colon Mode)

Extended Mode is used for save and quit or save without quit using "Esc" Key with ":"

Esc+:w	To Save the changes	j"-'\'\" .. /
Esc+:q	To quit (Without saving)	...-,-...
Esc+:wq	To save and quit	f \"
Esc+:w!	To save forcefully	/ _-\" \.>-:/
Esc+wq!	To save and quit forcefully	f_ \" 7
Esc+:x	To save and quit	
Esc+:X	To give passw or d to the file and r emove password	
Esc+:20(n)	To go to line no 20 or n	
Esc+: se nu	To set the line numbers to the file	!
Esc+:se nonu	To Remove the set line numbers	

Is command options

Options	Description
-l	Long listing format of files and directories, one per line
-a	List all hidden files and directories started with '.'
-F	Add a '/' classification at the end of each Directory
-g	List all files and directories with the group name
-i	Print index number of each files and directories
-m	List all file and directories separated by comma ','
-n	List numeric UID and GID of Owner and Groups
-r	List all files and directories in reverse order
-R	Short list all directories
-t	Sorted by modified time, started with the newest file

Types of files in linux.

File Type	First Character in File Listing	Description
Regular file	-	Normal files such as text, data, or executable files
Directory	d	Files that are lists of other files
Link	l	A shortcut that points to the location of the actual file
Special file	c	Mechanism used for input and output, such as files in /dev
Socket	s	A special file that provides inter-process networking protected by the file system's access control
Pipe	p	A special file that allows processes to communicate with each other without using network socket semantics

Symbolic links

Symbolic links are like desktop shortcuts we use in windows.

Create a soft link for /var/log directory in our current working directory.

```
imran@DevOps:~/linux-practices$ ls
file2  firstfile.txt  vmdir
imran@DevOps:~/linux-practices$ ls /var/log/
alternatives.log  auth.log.1  cups          fontconfig.log  kern.log.1    prime-supported.log  vbox-install.log  Xorg.1.log.old
alternatives.log.1  boot.log    dist-upgrade  fsck            lastlog       speech-dispatcher    wttmp             Xorg.2.log
appport.log        boot-sav    dmesg         gpu-manager.log  lightdm       syslog              wttmp.1
appport.log.1      bootstrap.log  dpkg.log     installer       old-logs      syslog.1            Xorg.0.log
apt               btmp        dpkg.log.1    jenkins         php7.0-fpm.log  unattended-upgrades  Xorg.0.log.old
auth.log           bttmp.1     faillog       kern.log        php7.0-fpm.log.1  upstart             Xorg.1.log
imran@DevOps:~/linux-practices$ ln -s /var/log/ logdir
imran@DevOps:~/linux-practices$ ls -l
total 8
-rw-rw-r-- 1 imran imran  0 Apr  2 18:00 file2
-rw-rw-r-- 1 imran imran 73 Apr  2 18:29 firstfile.txt
lrwxrwxrwx 1 imran imran  9 Apr  2 18:41 logdir -> /var/log/
drwxrwxr-x 4 imran imran 4096 Apr  2 18:21 vmdir
imran@DevOps:~/linux-practices$ ls
alternatives.log  auth.log.1  cups          fontconfig.log  kern.log.1    prime-supported.log  vbox-install.log  Xorg.1.log.old
alternatives.log.1  boot.log    dist-upgrade  fsck            lastlog       speech-dispatcher    wttmp             Xorg.2.log
appport.log        boot-sav    dmesg         gpu-manager.log  lightdm       syslog              wttmp.1
appport.log.1      bootstrap.log  dpkg.log     installer       old-logs      syslog.1            Xorg.0.log
apt               btmp        dpkg.log.1    jenkins         php7.0-fpm.log  unattended-upgrades  Xorg.0.log.old
auth.log           bttmp.1     faillog       kern.log        php7.0-fpm.log.1  upstart             Xorg.1.log
```


Note: -press **Enter** key to scroll down line by line (or)

Use **d** to go to next page

Use **b** to go to previous page

Use **/** to search for a word in the file

Use **v** to go vi mode where you can edit the file and once you save it you will back to less command

- more

more is exactly same like **less**

Ex: #more /etc/passwd

Note: -press **Enter** key to scroll down line by line (or)

Use **d** to go to next page

Use **/** to search for a word in the file

Use **v** to go vi mode where you can edit the file and once you save it you will back to more command

- head

It is used to display the top **10 lines** of the file.

Ex: # head /etc/passwd

```
[root@ktlinux ~]# head /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
uucp:x:10:14:uucp:/var/spool/uucp:/sbin/nologin
```

- tail

It is used to display the **last 10 lines** of the file

#tail /etc/passwd

```
[root@ktlinux ~]# tail /etc/passwd
apache:x:48:48:Apache:/var/www:/sbin/nologin
nslcd:x:65:55:LDAP Client User:/:/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/sbin/nologin
ntp:x:38:38:/etc/ntp:/sbin/nologin
pulse:x:496:494:PulseAudio System Daemon:/var/run/pulse:/sbin/nologin
gdm:x:42:42:/var/lib/gdm:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
tcpdump:x:72:72:/:/sbin/nologin
visitor:x:500:500:visitor:/home/visitor:/bin/bash
ktuser:x:501:501:/home/ktuser:/bin/bash
```


- cut

cut -d -f filename (where d stands for delimiter ex. ., " " etc and f stands for field)

```
[root@ktlinux ~]# cut -d: -f1 /etc/passwd
root
bin
daemon
adm
lp
sync
shutdown
halt
mail
uucp
```

To delimit spaces and print the field

#**cut -d " " -f1 filename**

- sed

sed stands for **s**tream **e**ditor, which is used to search a word in the file and replace it with the word required to be in the output

Note: it will only modify the output, but there will be no change in the original file.

#**sed 's/searchfor/replacewith/g' filename**

```
[root@ktlinux ~]# cat ktfile
Welcome to Kernel Tech
[root@ktlinux ~]# sed 's/Tech/Technologies/g' ktfile
Welcome to Kernel Technologies
[root@ktlinux ~]# cat ktfile
Welcome to Kernel Tech
```

I/O redirection

Redirection is a process where we can copy the output of any command(s), file(s) into a new file. There are two ways of redirecting the output into a file.

Using **>** or **>>** **filename** after the command, and

→ Create a file named devopstools with below content.

```
imran@DevOps:~/linux-practices$ cat devopstools
chef tech
ansible tech
git tech
docker tech
aws tech
```

→ Search for text “tech” replace it with “tools” and redirect output to a new file.

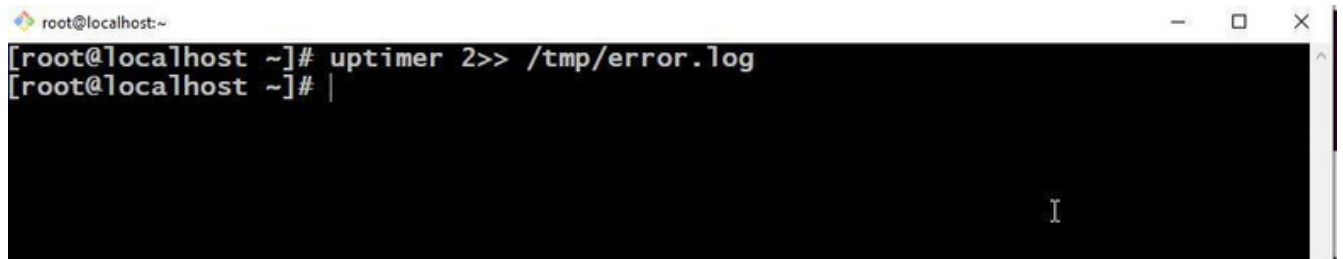
```
imran@DevOps:~/linux-practices$ sed 's/tech/tools/g' devopstools
chef tools
ansible tools
git tools
docker tools
aws tools
imran@DevOps:~/linux-practices$ sed 's/tech/tools/g' devopstools > newtools.txt
imran@DevOps:~/linux-practices$ cat newtools.txt
chef tools
ansible tools
git tools
docker tools
aws tools
```

Note: if the given name of the file is not available a new file will be created automatically. If the file already exists then it will overwrite contents of that file.

→ Appending another output in same file with “>>” .

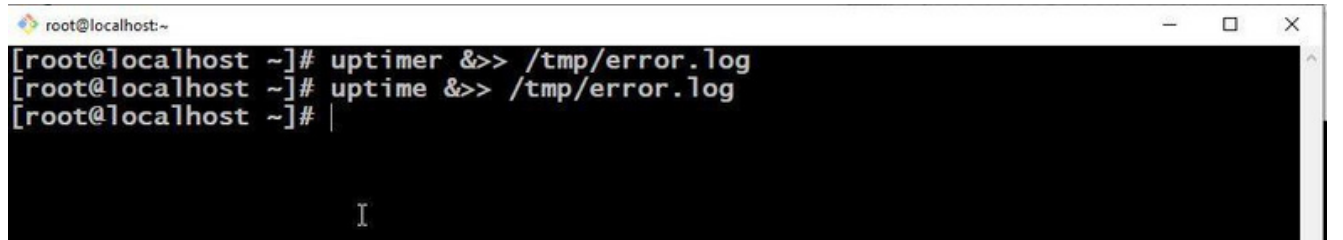
```
imran@DevOps:~/linux-practices$ tail /etc/passwd >> newtools.txt
imran@DevOps:~/linux-practices$ cat newtools.txt
chef tools
ansible tools
git tools
docker tools
aws tools
pulse:x:117:124:PulseAudio daemon,,,:/var/run/pulse:/bin/false
rtkit:x:118:126:RealtimeKit,,,:/proc:/bin/false
saned:x:119:127:./var/lib/saned:/bin/false
usbmux:x:120:46:usbmux daemon,,,:/var/lib/usbmux:/bin/false
imran:x:1000:1000:Imran,,,:/home/imran:/bin/bash
jenkins:x:121:131:Jenkins,,,:/var/lib/jenkins:/bin/bash
quest-lxlwni:x:999:999:Guest:/tmp/guest-lxlwni:/bin/bash
guests:x:1001:1001:./home/guests:/bin/bash
nvidia-persistenced:x:122:132:NVIDIA Persistence Daemon,,,:/sbin/nologin
quest-yjzlgk:x:998:998:Guest:/tmp/guest-yjzlgk:/bin/bash
imran@DevOps:~/linux-practices$
```

➔ Redirecting only error to a file “2>>”.

A terminal window titled 'root@localhost:~' with standard window controls. It shows the command 'uptimer 2>> /tmp/error.log' being executed, followed by a new prompt line '[root@localhost ~]# |' with a cursor at the end.

```
root@localhost:~  
[root@localhost ~]# uptimer 2>> /tmp/error.log  
[root@localhost ~]# |
```

➔ Redirecting all the output to a file “&>>”.

A terminal window titled 'root@localhost:~' with standard window controls. It shows two commands: 'uptimer &>> /tmp/error.log' and 'uptime &>> /tmp/error.log', followed by a new prompt line '[root@localhost ~]# |' with a cursor at the end.

```
root@localhost:~  
[root@localhost ~]# uptimer &>> /tmp/error.log  
[root@localhost ~]# uptime &>> /tmp/error.log  
[root@localhost ~]# |
```

Piping

So far we've dealt with sending data to and from files. Now we'll take a look at a mechanism for sending data from one program to another. It's called piping and the operator we use is (|). What this operator does is feed the output from the program on the left as input to the program on the right.

```
imran@DevOps:~$ cd linux-practices/
imran@DevOps:~/linux-practices$ ls
chefdk_1.2.22-1_amd64.deb  file2          logdir          tree_1.7.0-3_amd64.deb
devopstools               firstfile.txt  newtools.txt    vpdirt
imran@DevOps:~/linux-practices$ ls | head -3
chefdk_1.2.22-1_amd64.deb
devopstools
file2
imran@DevOps:~/linux-practices$ ls | grep logdir
logdir
imran@DevOps:~/linux-practices$ cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash
imran@DevOps:~/linux-practices$
```

Find

find command is used to find the files or directory's path, it is exactly like the find option in windows where you can search for a file.

```
imran@DevOps:~/linux-practices$ find /home/imran/ -name newtools.txt
/home/imran/linux-practices/newtools.txt
```

Options that can be used with find command:

Option	Usage
-name	For searching a file with its name
-inum	For searching a file with particular inode number
-type	For searching a particular type of file
-user	For files whose owner is a particular user
-group	For files belonging to particular group

5. Users & Groups.

USERS

Some Important Points related to Users:

- Users and groups are used to control access to files and resources
- Users login to the system by supplying their username and password
- Every file on the system is owned by a user and associated with a group
- Every process has an owner and group affiliation, and can only access the resources its owner or group can access.
- Every user of the system is assigned a unique user ID number (the UID)
- Users name and UID are stored in **/etc/passwd**
- User's password is stored in **/etc/shadow** in encrypted form.
- Users are assigned a **home directory** and a program that is run when they login (**Usually a shell**)
- Users cannot read, write or execute each other's files without permission.

Types of user

TYPE	EXAMPLE	USER ID (ID)	GROUP ID (GID)	HOME DIR	SHELL
ROOT	root	0	0	/root	/bin/bash
REGULAR	imran, vagrant	1000 to 60000	1000 to 60000	/home/username	/bin/bash
SERVICE	ftp, ssh, apache	1 to 999	1 to 999	/var/ftp etc	/sbin/nologin

In Linux there are three types of users.

1. Super user or root user

Super user or the root user is the most powerful user. He is the administrator user.

2. System user

System users are the users created by the softwares or applications. For example if we install Apache it will create a user apache. These kinds of users are known as system users.

3. Normal user

Normal users are the users created by root user. They are normal users like Rahul, Musab etc. Only the root user has the permission to create or remove a user.

Whenever a user is created in Linux things created by default:-

- A home directory is created(/home/username)
- A mail box is created(/var/spool/mail)
- unique UID & GID are given to user

Passwd file

1. /etc/passwd

```
[root@ktlinux ~]# head /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
```

The above fields are

- **root** = name
- **x** = link to password file i.e. /etc/shadow
- **0** or **1** = UID (user id)
- **0** or **1** = GID (group id)
- **root** or **bin** = comment (brief information about the user)
- **/root** or **/bin** = home directory of the user
- **/bin/bash** or **/sbin/nologin** = shell

Group file

2. /etc/group

The file /etc/group stores group information. Each line in this file stores one group entry.

Group name, group password, GID, group members

```
[root@localhost ~]# head /etc/group
root:x:0:
bin:x:1:
daemon:x:2:
```

ADD USER, SET PASSWORD & SWITCH TO USER

```
dino@localhost:~  
[vagrant@localhost ~]$ sudo useradd dino  
[vagrant@localhost ~]$ sudo passwd dino  
Changing password for user dino.  
New password:  
Retype new password:  
Sorry, passwords do not match.  
New password:  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[vagrant@localhost ~]$ su - dino  
Password:  
[dino@localhost ~]$ pwd  
/home/dino  
[dino@localhost ~]$ id  
uid=1002(dino) gid=1003(dino) groups=1003(dino) context=unconfined_u:unconfined_  
r:unconfined_t:s0-s0:c0.c1023  
[dino@localhost ~]$ |
```

ADD USER, GROUP & USER INTO GROUP

```
root@localhost:~  
[root@localhost ~]# useradd devops  
[root@localhost ~]# id devops  
uid=1001(devops) gid=1001(devops) groups=1001(devops)  
[root@localhost ~]# grep devops /etc/passwd  
devops:x:1001:1001::/home/devops:/bin/bash  
[root@localhost ~]# groupadd opsadmin  
[root@localhost ~]# usermod -G opsadmin devops  
[root@localhost ~]# grep opsadmin /etc/group  
opsadmin:x:1002:devops  
[root@localhost ~]# id devops  
uid=1001(devops) gid=1001(devops) groups=1001(devops),1002(opsadmin)  
[root@localhost ~]# |
```

DELETE USER & GROUP

```
vagrant@localhost:~  
[vagrant@localhost ~]$ sudo userdel -r dino  
[vagrant@localhost ~]$ sudo groupdel opsadmin  
[vagrant@localhost ~]$ sudo id dino  
id: dino: no such user  
[vagrant@localhost ~]$ |
```

3. The /etc/shadow file

This file stores users' password and password related information. Just like */etc/passwd* file, this file also uses an individual line for each entry.

1. Username
2. Encrypted password
3. Number of days when password was last changed
4. Number of days before password can be changed
5. Number of days after password must be changed
6. Number of days before password expiry date to display the warning message

7. Number of days to disable the account after the password expiry
8. Number of days since the account is disabled
9. Reserved field

```
[root@localhost ~]# cat /etc/shadow
root:$1$m.FEVNiS$OYiaRNHMHZS85/wnDHccI.:0
bin:!:18353:0:99999:7:::
daemon:!:18353:0:99999:7:::
adm:!:18353:0:99999:7:::
lp:!:18353:0:99999:7:::
sync:!:18353:0:99999:7:::
shutdown:!:18353:0:99999:7:::
halt:!:18353:0:99999:7:::
mail:!:18353:0:99999:7:::
```

USER & GROUP cheatsheet

COMMANDS	DESCRIPTION
Creates user in RedHat	
useradd	Creates user in ubuntu Shows user info
adduser	Creates group Adds user to group
id	set/reset password removes user with
groupadd	home dir removes group shows last login
usermod -G grpnam username	in system who is logged into system
passwd	username
userdel -r	List files opened by user
groupdel	
last	
who	
whoami	
lsof -u user	

6. File permissions

Viewing Permissions from the Command-Line

- File permissions may be viewed using **ls -l**

```
$ ls -l /bin/login  
-rwxr-xr-x 1 root root 19080 Apr 1 18:26 /bin/login
```

- Four symbols are used when displaying permissions:
 - r: permission to read a file or list a directory's contents
 - w: permission to write to a file or create and remove files from a directory
 - x: permission to execute a program or change into a directory and do a long listing of the directory
 - -: no permission (in place of the r, w, or x)

Changing File Ownership

- Only root can change a file's owner
- Only root or the owner can change a file's group
- Ownership is changed with **chown**:
 - **chown [-R] user_name file|directory ...**
- Group-Ownership is changed with **chgrp**:
 - **chgrp [-R] group_name file|directory ...**


```

imran@DevOps:~/linux-practices$ sudo adduser sam
[sudo] password for imran:
Adding user 'sam' ...
Adding new group 'sam' (1002) ...
Adding new user 'sam' (1002) with group 'sam' ...
Creating home directory '/home/sam' ...
Copying files from '/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for sam
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
imran@DevOps:~/linux-practices$ id sam
uid=1002(sam) gid=1002(sam) groups=1002(sam)
imran@DevOps:~/linux-practices$ ls -l devopstools
-rw-rw-r-- 1 imran imran 53 Apr  2 19:09 devopstools
imran@DevOps:~/linux-practices$ chown sam.sam devopstools
chown: changing ownership of 'devopstools': Operation not permitted
imran@DevOps:~/linux-practices$ sudo chown sam.sam devopstools
imran@DevOps:~/linux-practices$ ls -l devopstools
-rw-rw-r-- 1 sam sam 53 Apr  2 19:09 devopstools
imran@DevOps:~/linux-practices$ ls
devopstools  file2  firstfile.txt  logdir  newtools.txt  vpdire
imran@DevOps:~/linux-practices$ ls -l vpdire/
total 8
drwxrwxr-x 4 imran imran 4096 Apr  2 18:14 devopsdire
-rw-rw-r-- 1 imran imran   0 Apr  2 18:00 file3
-rw-rw-r-- 1 imran imran   0 Apr  2 18:00 file4
drwxrwxr-x 2 imran imran 4096 Apr  2 18:17 testdire
imran@DevOps:~/linux-practices$ chown sam.sam vpdire/ -R
chown: changing ownership of 'vpdire/devopsdire/ansible': Operation not permitted
chown: changing ownership of 'vpdire/devopsdire/aws': Operation not permitted
chown: changing ownership of 'vpdire/devopsdire': Operation not permitted
chown: changing ownership of 'vpdire/file4': Operation not permitted
chown: changing ownership of 'vpdire/file3': Operation not permitted

```

Changing Permissions - Symbolic Method

- To change access modes:
 - o `chmod [-OPTION] ... mode[,mode] file directory ...`
- *mode* includes:
 - o `u,g,o` for user, group and other
 - o `+` - `OR` for grant, deny or set
 - o `r, w or x` for read, write and execute
- Options include:
 - o `-R` Recursive
 - o `-V` Verbose
 - o `--reference` Reference another file for its mode
- Examples:
 - o `chmod ugo+r file:` Grant read access to all for *file*
 - o `chmod o-wx dire:` Deny write and execute to others for *dir*

Changing Permissions - Numeric Method

- Uses a three-digit mode number
 - first digit specifies owner 's permissions
 - second digit specifies group permissions
 - third digit represents others' permissions
- Permissions are calculated by adding:
 - 4 (for read)
 - 2 (for write)
 - 1 (for execute)
- Example:
 - `chmod 640 myfile`

```
imran@DevOps:~/linux-practices$ ls -l
total 16
-rw-rw-r-- 1 sam sam 53 Apr 2 19:09 devopstools
-rw-rw-r-- 1 imran imran 0 Apr 2 18:00 file2
-rw-rw-r-- 1 imran imran 73 Apr 2 18:29 firstfile.txt
lrwxrwxrwx 1 imran imran 9 Apr 2 18:41 logdir -> /var/log/
-rw-rw-r-- 1 imran imran 612 Apr 2 19:14 newtools.txt
drwxrwxr-x 4 sam sam 4096 Apr 2 18:21 vpdire
imran@DevOps:~/linux-practices$ chmod u+x newtools.txt
imran@DevOps:~/linux-practices$ ls -l newtools.txt
-rwxrw-r-- 1 imran imran 612 Apr 2 19:14 newtools.txt
imran@DevOps:~/linux-practices$ chmod o-r newtools.txt
imran@DevOps:~/linux-practices$ ls -l newtools.txt
-rwxrw---- 1 imran imran 612 Apr 2 19:14 newtools.txt
imran@DevOps:~/linux-practices$ chmod 700 newtools.txt
imran@DevOps:~/linux-practices$ ls -l newtools.txt
-rwx----- 1 imran imran 612 Apr 2 19:14 newtools.txt
imran@DevOps:~/linux-practices$ chmod 755 newtools.txt
imran@DevOps:~/linux-practices$ ls -l newtools.txt
-rwxr-xr-x 1 imran imran 612 Apr 2 19:14 newtools.txt
imran@DevOps:~/linux-practices$
```


7. Sudo

sudo gives power to a normal user to execute commands which is owned by root user.

Example shown below:

If a user has already full sudoers privilege, it can become a root user anytime.

→ sudo -i changes from normal user to root user

```
imran@DevOps:~/linux-practices$ id
uid=1000(imran) gid=1000(imran) groups=1000(imran),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),113(lpadmin),128(sambashare),130(docker),1006(dev_dock_gr)
imran@DevOps:~/linux-practices$ sudo -i
[sudo] password for imran:
root@DevOps:~# id
uid=0(root) gid=0(root) groups=0(root)
root@DevOps:~#
```

Note: User imran was already a sudo user with full privilege.

→ Adding user sam in sudoers list.

```
imran@DevOps:~/linux-practices$ sudo -i
root@DevOps:~# export EDITOR=vim
root@DevOps:~# visudo
```

```
File Edit View Search Terminal Help
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
sam     ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL
%dev_dock ALL=(ALL:ALL) ALL
```

21,3 Top

→ Like a user a group can also be added into sudoers list.

```
# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL
```

- Every time you enter sudo command it asks your own password. To turn that off use NOPASSWD in sudoers file.

```
# User privilege specification
root    ALL=(ALL:ALL) ALL
sam     ALL=(ALL:ALL) NOPASSWD: ALL
```

- Changing to any other user with “su -” command.

```
imran@DevOps:~/linux-practices$ su - sam
Password:
sam@DevOps:~$
```

- Become a root user from sam user login.

```
sam@DevOps:~$ sudo -i
root@DevOps:~#
```

8. Software Management.

→ Download package from internet.

For CentOS

To install Tree

```
# curl https://rpmfind.net/linux/centos/7.9.2009/os/x86_64/Packages/tree-1.6.0-10.el7.x86_64.rpm -o tree-1.6.0-10.el7.x86_64.rpm
```

```
# rpm -ivh tree-1.6.0-10.el7.x86_64.rpm
```

```
[root@Imran ~]# curl https://rpmfind.net/linux/centos/7.9.2009/os/x86_64/Packages/tree-1.6.0-10.el7.x86_64.rpm -o tree-1.6.0-10.el7.x86_64.rpm
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 47508  100 47508    0     0  57226      0  0 --:--:-- --:--:-- --:--:-- 57238
[root@Imran ~]# ls
anaconda-ks.cfg  httpd-2.4.6-95.el7.centos.x86_64.rpm  original-ks.cfg  tree-1.6.0-10.el7.x86_64.rpm
[root@Imran ~]# rpm -ivh tree-1.6.0-10.el7.x86_64.rpm
Preparing...                ##### [100%]
Updating / installing...
 1:tree-1.6.0-10.el7        ##### [100%]
[root@Imran ~]# tree /var/log/
/var/log/
├── anaconda
│   ├── anaconda.log
│   ├── ifcfg.log
│   ├── journal.log
│   ├── ks-script-898Bxq.log
│   ├── ks-script-dyarrY.log
│   ├── ks-script-kPd16m.log
│   ├── ks-script-wnz4e2.log
│   ├── packaging.log
│   ├── program.log
│   ├── storage.log
│   └── syslog
├── audit
│   └── audit.log
├── boot.log
├── btmp
├── chrony
├── cloud-init.log
├── cron
├── dmesg
├── grubby_prune_debug
├── lastlog
├── maillog
├── messages
├── qemu-ga
├── rhsm
├── secure
└── spooler
```

To install httpd

```
# curl https://rpmfind.net/linux/centos/7.9.2009/os/x86_64/Packages/httpd-2.4.6-95.el7.centos.x86_64.rpm -o 2.4.6-95.el7.centos.x86_64.rpm
```

```
# rpm -ivh httpd-2.4.6-95.el7.centos.x86_64.rpm
```

```
[root@Imran ~]# curl https://rpmfind.net/linux/centos/7.9.2009/os/x86_64/Packages/httpd-2.4.6-95.el7.centos.x86_64.rpm -o httpd-2.4.6-95.el7.centos.x86_64.rpm
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 2779k  100 2779k    0     0 1943k      0  0 0:00:01 0:00:01 --:--:-- 1945k
[root@Imran ~]# ls
anaconda-ks.cfg  httpd-2.4.6-95.el7.centos.x86_64.rpm  original-ks.cfg
[root@Imran ~]# rpm -ivh httpd-2.4.6-95.el7.centos.x86_64.rpm
warning: httpd-2.4.6-95.el7.centos.x86_64.rpm: Header V3 RSA/SHA256 Signature, key ID f4a80eb5: NOKEY
error: Failed dependencies:
  /etc/mime.types is needed by httpd-2.4.6-95.el7.centos.x86_64
  system-logos >= 7.92.1-1 is needed by httpd-2.4.6-95.el7.centos.x86_64
  httpd-tools = 2.4.6-95.el7.centos is needed by httpd-2.4.6-95.el7.centos.x86_64
  libapr-1.so.0(64bit) is needed by httpd-2.4.6-95.el7.centos.x86_64
  libaprutil-1.so.0(64bit) is needed by httpd-2.4.6-95.el7.centos.x86_64
[root@Imran ~]#
```

Due to Dependencies its failing and it will be installed one we install all the dependencies. But what if we have Hundreds of dependencies, And that can be solved easily by other package managers like YUM .

repos. d/ directory. It reads each YUM Repository configuration file to get the information required to

download and install new software, resolves software dependencies and installs the required RPM pack files. YUM Repository configuration files must: be located in /etc/yum.repos.d

```
# ls /etc/yum.repos.d/
```

```
[root@Imran ~]#  
[root@Imran ~]# ls /etc/yum.repos.d/  
CentOS-Base.repo  CentOS-CR.repo  CentOS-Debuginfo.repo  CentOS-fasttrack.repo  CentOS-Media.repo  CentOS-Sources.repo  CentOS-Vault.repo
```

Shows the usage of YUM Command with options

```
# yum --help
```

```
[root@Imran ~]#  
[root@Imran ~]# yum --help  
Loaded plugins: fastestmirror  
Usage: yum [options] COMMAND  
  
List of Commands:  
  
check           Check for problems in the rpmdb  
check-update    Check for available package updates  
clean           Remove cached data  
deplist         List a package's dependencies  
distribution-synchronization Synchronize installed packages to the latest available versions  
downgrade       downgrade a package  
erase           Remove a package or packages from your system  
fs              Acts on the filesystem data of the host, mainly for removing docs/languages for minimal hosts.  
fssnapshot      Creates filesystem snapshots, or lists/deletes current snapshots.  
groups          Display, or use, the groups information  
help            Display a helpful usage message  
history         Display, or use, the transaction history  
info            Display details about a package or group of packages  
install         Install a package or packages on your system  
list            List a package or groups of packages  
load-transaction load a saved transaction from filename  
makecache       Generate the metadata cache  
provides        Find what package provides the given value  
reinstall       reinstall a package  
repo-pkgs       Treat a repo. as a group of packages, so we can install/remove all of them  
repolist        Display the configured software repositories  
search          Search package details for the given string  
shell           Run an interactive yum shell  
swap            Simple way to swap packages, instead of using shell  
update          Update a package or packages on your system  
update-minimal  Works like upgrade, but goes to the 'newest' package match which fixes a problem that affects your system  
updateinfo      Acts on repository update information  
upgrade         Update packages taking obsoletes into account  
version         Display a version for the machine and/or available repos.  
  
Options:  
-h, --help          show this help message and exit  
-t, --tolerant       be tolerant of errors
```

To Update all your packages

```
#yum update
```

```
[root@Imran ~]# yum update  
Loaded plugins: fastestmirror  
Loading mirror speeds from cached hostfile  
* base: download.cf.centos.org  
* extras: download.cf.centos.org  
* updates: download.cf.centos.org  
Resolving Dependencies  
--> Running transaction check  
--> Package acl.x86_64 0:2.2.51-14.el7 will be updated  
--> Package acl.x86_64 0:2.2.51-15.el7 will be an update  
--> Package bash.x86_64 0:4.2.46-33.el7 will be updated  
--> Package bash.x86_64 0:4.2.46-34.el7 will be an update  
--> Package bind-export-libs.x86_64 32:9.11.4-9.P2.el7 will be updated  
--> Package bind-export-libs.x86_64 32:9.11.4-26.P2.el7_9.7 will be an update  
--> Package binutils.x86_64 0:2.27-41.base.el7_7.2 will be updated  
--> Package binutils.x86_64 0:2.27-44.base.el7 will be an update  
--> Package ca-certificates.noarch 0:2019.2.32-76.el7_7 will be updated  
--> Package ca-certificates.noarch 0:2021.2.50-72.el7_9 will be an update  
--> Package centos-release.x86_64 0:7-7.1908.0.el7.centos will be updated  
--> Package centos-release.x86_64 0:7-9.2009.1.el7.centos will be an update  
--> Package chkconfig.x86_64 0:1.7.4-1.el7 will be updated  
--> Package chkconfig.x86_64 0:1.7.6-1.el7 will be an update  
--> Package cloud-init.x86_64 0:18.5-3.el7.centos will be updated
```

To install httpd

```
# yum install httpd -y
```



```

[root@Imran ~]#
[root@Imran ~]# yum install httpd -y
Loaded plugins: fastestmirror
Determining fastest mirrors
 * base: download.cf.centos.org
 * extras: download.cf.centos.org
 * updates: download.cf.centos.org
base | 3.6 kB 00:00:00
extras | 2.9 kB 00:00:00
updates | 2.9 kB 00:00:00
(1/4): base/7/x86_64/group_gz | 153 kB 00:00:00
(2/4): extras/7/x86_64/primary_db | 243 kB 00:00:00
(3/4): base/7/x86_64/primary_db | 6.1 MB 00:00:00
(4/4): updates/7/x86_64/primary_db | 11 MB 00:00:00
Resolving Dependencies
--> Running transaction check
Total | 22 MB/s | 24 MB 00:00:01
Retrieving key from file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
Importing GPG key 0xF4A80EB5:
 Userid : "CentOS-7 Key (CentOS 7 Official Signing Key) <security@centos.org>"
 Fingerprint: 6341 ab27 53d7 8a78 a7c2 7bb1 24c6 a8a7 f4a8 0eb5
 Package : centos-release-7-7.1908.0.el7.centos.x86_64 (installed)
 From : /etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Installing : apr-1.4.8-7.el7.x86_64 1/6
Installing : apr-util-1.5.2-6.el7.x86_64 2/6
Installing : httpd-tools-2.4.6-97.el7.centos.x86_64 3/6
Installing : centos-logos-70.0.6-3.el7.centos.noarch 4/6
Installing : mailcap-2.1.41-2.el7.noarch 5/6
Installing : httpd-2.4.6-97.el7.centos.x86_64 6/6
Verifying : mailcap-2.1.41-2.el7.noarch 1/6
Verifying : apr-1.4.8-7.el7.x86_64 2/6
Verifying : apr-util-1.5.2-6.el7.x86_64 3/6
Verifying : httpd-2.4.6-97.el7.centos.x86_64 4/6
Verifying : httpd-tools-2.4.6-97.el7.centos.x86_64 5/6
Verifying : centos-logos-70.0.6-3.el7.centos.noarch 6/6

Installed:
httpd.x86_64 0:2.4.6-97.el7.centos

Dependency Installed:
apr.x86_64 0:1.4.8-7.el7 apr-util.x86_64 0:1.5.2-6.el7 centos-logos.noarch 0:70.0.6-3.el7.centos httpd-tools.x86_64 0:2.4.6-97.el7.centos
mailcap.noarch 0:2.1.41-2.el7

Complete!

```

To remove httpd

#yum remove httpd -y

```

[root@Imran ~]# yum remove httpd -y
Loaded plugins: fastestmirror
Resolving Dependencies
--> Running transaction check
--> Package httpd.x86_64 0:2.4.6-97.el7.centos will be erased
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package Arch Version Repository Size
=====
Removing:
httpd x86_64 2.4.6-97.el7.centos @updates 9.4 M
=====

Transaction Summary
=====
Remove 1 Package

Installed size: 9.4 M
Downloading packages:
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Erasing : httpd-2.4.6-97.el7.centos.x86_64 1/1
Verifying : httpd-2.4.6-97.el7.centos.x86_64 1/1

Removed:
httpd.x86_64 0:2.4.6-97.el7.centos

Complete!

```

For Ubuntu

wget http://archive.ubuntu.com/ubuntu/pool/universe/t/tree/tree_1.7.0-3_amd64.deb -o tree_1.7.0-3_amd64.deb

dpkg -i tree_1.7.0-3_amd64.deb


```

root@Imran:~#
root@Imran:~# wget http://archive.ubuntu.com/ubuntu/pool/universe/t/tree/tree_1.7.0-3_amd64.deb
--2021-09-27 04:30:17-- http://archive.ubuntu.com/ubuntu/pool/universe/t/tree/tree_1.7.0-3_amd64.deb
Resolving archive.ubuntu.com (archive.ubuntu.com)... 91.189.88.142, 91.189.88.152, 2001:67c:1360:8001::24, ...
Connecting to archive.ubuntu.com (archive.ubuntu.com)|91.189.88.142|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 40572 (40K) [application/x-debian-package]
Saving to: 'tree_1.7.0-3_amd64.deb'

tree_1.7.0-3_amd64.deb      100%[=====>] 39.62K  --.-KB/s   in 0.07s

2021-09-27 04:30:17 (551 KB/s) - 'tree_1.7.0-3_amd64.deb' saved [40572/40572]

root@Imran:~# dpkg -i tree_1.7.0-3_amd64.deb
Selecting previously unselected package tree.
(Reading database ... 63739 files and directories currently installed.)
Preparing to unpack tree_1.7.0-3_amd64.deb ...
Unpacking tree (1.7.0-3) ...
Setting up tree (1.7.0-3) ...
Processing triggers for man-db (2.9.1-1) ...
root@Imran:~# tree /var/log/
/var/log/
├── amazon
│   ├── ssm
│   │   ├── amazon-ssm-agent.log
│   │   ├── audits
│   │   └── amazon-ssm-agent-audit-2021-09-27
│   └── hibernate.log
├── apt
│   ├── eipp.log.xz
│   ├── history.log
│   └── term.log
├── auth.log
├── btmp
├── cloud-init-output.log
├── cloud-init.log
├── dist-upgrade
└── dmesg

```

We have seen YUM Like that for Ubuntu we have a package manager ‘apt’.

The sources. list file is a key factor in adding or upgrading applications to your Ubuntu installation. This is also used by your system for system updates. The file is basically the roadmap for your system to know what it may download programs for installation or upgrade.

cat /etc/apt/sources.list

```

root@Imran:~#
root@Imran:~# cd /etc/apt/
root@Imran:/etc/apt# ls
apt.conf.d  auth.conf.d  preferences.d  sources.list  sources.list.d  trusted.gpg.d
root@Imran:/etc/apt# cat /etc/apt/sources.list
## Note, this file is written by cloud-init on first boot of an instance
## modifications made here will not survive a re-bundle.
## if you wish to make changes you can:
## a.) add 'apt_preserve_sources_list: true' to /etc/cloud/cloud.cfg
##    or do the same in user-data
## b.) add sources in /etc/apt/sources.list.d
## c.) make changes to template file /etc/cloud/templates/sources.list.tpl

# See http://help.ubuntu.com/community/UpgradeNotes for how to upgrade to
# newer versions of the distribution.
deb http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal main restricted
# deb-src http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal main restricted

## Major bug fix updates produced after the final release of the
## distribution.
deb http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal-updates main restricted
# deb-src http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal-updates main restricted

## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team. Also, please note that software in universe WILL NOT receive any
## review or updates from the Ubuntu security team.
deb http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal universe
# deb-src http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal universe
deb http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal-updates universe
# deb-src http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal-updates universe

## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team, and may not be under a free licence. Please satisfy yourself as to
## your rights to use the software. Also, please note that software in
## multiverse WILL NOT receive any review or updates from the Ubuntu
## security team.
deb http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal multiverse
# deb-src http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal multiverse
deb http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal-updates multiverse
# deb-src http://us-east-1.ec2.archive.ubuntu.com/ubuntu/ focal-updates multiverse

```

Shows the usage of apt Command with options

apt --help

```

root@Imran:~# apt --help
apt 2.0.6 (amd64)
Usage: apt [options] command

apt is a commandline package manager and provides commands for
searching and managing as well as querying information about packages.
It provides the same functionality as the specialized APT tools,
like apt-get and apt-cache, but enables options more suitable for
interactive use by default.

Most used commands:
  list - list packages based on package names
  search - search in package descriptions
  show - show package details
  install - install packages
  reinstall - reinstall packages
  remove - remove packages
  autoremove - Remove automatically all unused packages
  update - update list of available packages
  upgrade - upgrade the system by installing/upgrading packages
  full-upgrade - upgrade the system by removing/installing/upgrading packages
  edit-sources - edit the source information file
  satisfy - satisfy dependency strings

See apt(8) for more information about the available commands.
Configuration options and syntax is detailed in apt.conf(5).
Information about how to configure sources can be found in sources.list(5).
Package and version choices can be expressed via apt_preferences(5).
Security details are available in apt-secure(8).

This APT has Super Cow Powers.

```

To update all your package lists

#apt update

```

root@Imran:~# apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu focal-security InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
27 packages can be upgraded. Run 'apt list --upgradable' to see them.

```

TO search for a <package> apache2

apt search apache2

```

root@Imran:~# apt search apache2
Sorting... Done
Full Text Search... Done
apache2/focal-updates,now 2.4.41-4ubuntu3.4 amd64 [installed]
  Apache HTTP Server

apache2-bin/focal-updates,now 2.4.41-4ubuntu3.4 amd64 [installed,automatic]
  Apache HTTP Server (modules and other binary files)

apache2-data/focal-updates,now 2.4.41-4ubuntu3.4 all [installed,automatic]
  Apache HTTP Server (common files)

apache2-dev/focal-updates 2.4.41-4ubuntu3.4 amd64
  Apache HTTP Server (development headers)

apache2-doc/focal-updates 2.4.41-4ubuntu3.4 all
  Apache HTTP Server (on-site documentation)

apache2-ssl-dev/focal-updates 2.4.41-4ubuntu3.4 amd64
  Apache HTTP Server (mod_ssl development headers)

apache2-suexec-custom/focal-updates 2.4.41-4ubuntu3.4 amd64
  Apache HTTP Server configurable suexec program for mod_suexec

apache2-suexec-pristine/focal-updates 2.4.41-4ubuntu3.4 amd64
  Apache HTTP Server standard suexec program for mod_suexec

apache2-utils/focal-updates,now 2.4.41-4ubuntu3.4 amd64 [installed,automatic]
  Apache HTTP Server (utility programs for web servers)

```

To install apache2

apt install apache2 -y

```

root@Imran:~# apt install apache2 -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2
0 upgraded, 1 newly installed, 0 to remove and 27 not upgraded.
Need to get 95.5 kB of archives.
After this operation, 542 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 apache2 amd64 2.4.41-4ubuntu3.4 [95.5 kB]
Fetched 95.5 kB in 0s (5456 kB/s)
Selecting previously unselected package apache2.
(Reading database ... 64421 files and directories currently installed.)
Preparing to unpack .../apache2_2.4.41-4ubuntu3.4_amd64.deb ...
Unpacking apache2 (2.4.41-4ubuntu3.4) ...
Setting up apache2 (2.4.41-4ubuntu3.4) ...
Processing triggers for systemd (245.4-4ubuntu3.11) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ufw (0.36-6) ...
root@Imran:~#

```

To remove apache2

apt remove apache2 -y

```

root@Imran:~# apt remove apache2 -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.2-0 ssl-cert
Use 'apt autoremove' to remove them.
The following packages will be REMOVED:
  apache2
0 upgraded, 0 newly installed, 1 to remove and 27 not upgraded.
After this operation, 542 kB disk space will be freed.
(Reading database ... 64471 files and directories currently installed.)
Removing apache2 (2.4.41-4ubuntu3.4) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ufw (0.36-6) ...
root@Imran:~#

```

Commands	Description	
wget link	to download file from link	
curl link	access file from link	
curl link -o outputfile	access file and store o/p to a file	
REDHAT RPM commands		
rpm -ivh {rpm-file}	Install the package	rpm -ivh mozilla-mail-1.7.5-17.i586.rpm rpm -ivh --test mozilla-mail-1.7.5-17.i586.rpm
rpm -Uvh {rpm-file}	Upgrade package	rpm -Uvh mozilla-mail-1.7.6-12.i586.rpm rpm -Uvh --test mozilla-mail-1.7.6-12.i586.rpm

rpm -ev {package}	Erase/remove/ an installed package	rpm -ev mozilla-mail
rpm -ev --nodeps {package}	Erase/remove/ an installed package without checking for	rpm -ev --nodeps mozilla-mail

	dependencies	
rpm -qa	Display list all installed rpm packages Display installed	rpm -qa less
rpm -qi {package}	information along with package version and short description Find out what package	rpm -qi mozilla-mail rpm -qf /etc/passwd
rpm -qf {/path/to/file}	a file belongs to i.e. find what package owns the file	rpm -qf /bin/bash
rpm -qc {package-name}	Display list of configuration file(s) for a package	rpm -qc httpd
rpm -qcf {/path/to/file}	Display list of configuration files for a command	rpm -qcf /usr/X11R6/bin/xeyes
rpm -qa --last	Display list of all recently installed RPMs	rpm -qa --last rpm -qa --last less
rpm -qpR {rpm-file} rpm -qR {package}	Find out what dependencies a rpm file has	rpm -qpR mediawiki-1.4rc1-4.i586.rpm rpm -qR bash
CentOS_8 Commands		
DNF commands cheatsheet	https://www.linuxtechi.com/dnf-command-examples-rpm-management-fedora-linux/	
dnf --help	Show the help	

<code>dnf search PACKAGE</code>	search from available repositories	
<code>dnf install PACKAGE -y</code>	To install the package	
<code>dnf install httpd -y</code>	To Install httpd package	
<code>dnf install vim -y</code>	Installing VIM Editor	
<code>dnf reinstall PACKAGE</code>	To reinstall PACKAGE	
<code>dnf remove PACKAGE</code>	To remove PACKAGE	
<code>dnf update</code>	update all packages	
<code>dnf update PACKAGE</code>	update only a package	

<code>dnf grouplist</code>	List all available Group Packages	
<code>dnf groupinstall "GROUPNAME"</code>	Installs all the packages in a group	
<code>dnf repolist</code>	List Enabled dnf Repositories	
<code>dnf clean all</code>	Clean dnf Cache	
<code>dnf install epel-release</code>	Additional package repository that provides easy access to install packages for commonly used software.	

dnf history	View History of dnf	
dnf info package name	Shows the information of package like version, size, source, repository etc	
YUM Commands Cheatsheet	https://access.redhat.com/sites/default/files/attachments/rh_yum_cheatsheet_1214_jcs_print-1.pdf	
yum -help	Shows the help	
yum search PACKAGE	Search from available repositories	
yum install PACKAGE -y	To install the package	
yum install httpd -y	To install httpd package	
yum install vim -y	To install VIM Editor	
yum reinstall PACKAGE	To reinstall the PACKAGE	
Yum remove PACKAGE	To Remove PACKAGE	
yum update	Update all packages	
yum update PACKAGE	To Update specific package	
yum grouplist	List all available Group packages	
yum groupinstall "Group Name"	Install all the packages in a group	

Yum repolist	List Enabled YUM repositories	
yum install epel-release	Additional package repository that provides easy access to install packages for commonly used software.	
yum clean all	Clean yum cache	
yum history	View history of yum	
Yum info PACKAGE NAME	Shows the information of package like version, size, source, repository etc.	
Ubuntu20 Commands		
apt commands cheatsheet	https://itsfoss.com/apt-command-guide/	
apt search PACKAGE	search from available repositories	
apt install PACKAGE -y	To Install Packages	
apt install apache2 -y	To Install apache2	
apt reinstall PACKAGE	To reinstall PACKAGE	
apt remove PACKAGE	To remove PACKAGE	
apt update	update all packages	
apt update PACKAGE	update only a package	

apt grouplist	List all available Group Packages -----	
apt groupinstall "GROUPNAME"	Installs all the packages in a group.	
apt repolist	List Enabled apt Repositories -----	
apt clean all	Clean apt Cache	
apt history	View History of apt -----	
apt show package name	Shows the information of package like version, size, source, repository etc	

9. SEARCH

```
$ grep pattern files
(you will this command
often)
$ grep -r pattern dir
pattern in dir

$ locate
file
$ find /home/tom -name
'index*' with "index"(10 find
examples)
$ find /home -size
+10000k 10000k in /home

# Search for pattern in
files
# Search recursively
for
# Find all instances of
file
# Find files names that
start
# Find files larger
than
```

10. LOGIN (SSH AND TELNET)

```
$ ssh user@host
(secure data communication
command)
$ ssh -p port
user@host specific
port
$ telnet host
using telnet
port

# Connect to host as
user
# Connect to host
using
# Connect to the
system
```

11. FILE TRANSFER

```
scp

$ scp file.txt
file.txt to remote host /tmp folder

# Secure

$ scp nixsavy@server2:/www/*.html /www/tmp # Copy *.html
files from remote host to current system /www/tmp folder

$ scp -r nixsavy@server2:/www /www/tmp # Copy all
and folders recursively from remote server to the current
system
/www/tmp folder

rsync

$ rsync -a /home/apps
/backup/ source to
destination

#
Synchronize

$ rsync -avz /home/apps
Synchronize files/directories between the local and remote
system with compression enabled
```

12. DISK USAGE

```
$ df # Show free space on filesystems (commonly used command)
$ df -i # Show free inodes on filesystems
$ fdisk -l # Show disks partitions sizes and types (fdisk command output)
$ du # Display disk usage in readable form (command variations)
$ du -sh # Display total disk usage on current directory
$ # Displays target mount point all filesystem (refer type,list,evaluate output)
$ mount device-path mount-point # Mount a device
```

13. DIRECTORY TRAVERSE

```
$ cd .. # To go up one level of the directory tree (simple & most needed)
$ # Go to $HOME directory
$ cd /test # Change to /test directory
```

14. SERVICES

Centos8

<i>\$ sudo systemctl start httpd</i>	<i># Starts httpd on centos</i>
<i>\$ sudo systemctl stop httpd</i>	<i># Stops httpd on centos</i>
<i>\$ sudo systemctl restart httpd</i>	<i># Restart services</i>
<i>\$ sudo systemctl status httpd</i>	<i># shows the current status</i>
<i>\$ sudo systemctl reload httpd</i>	<i># Reload conf</i>
<i>\$ sudo systemctl enable httpd</i>	<i># starts httpd at boot time</i>
<i>\$ sudo systemctl disable httpd</i>	<i># stops httpd at boot time</i>

<code>\$ sudo systemctl is-active httpd</code>	<code># shows whether the service is active or not</code>
<code>\$ sudo systemctl is-enabled httpd</code>	<code># shows whether the service is enabled or not</code>

Ubuntu20

<code>\$ sudo systemctl start apache2</code>	<code># Starts apache2 on ubuntu</code>
<code>\$ sudo systemctl stop apache2</code>	<code># Stops apache2 on ubuntu</code>
<code>\$ sudo systemctl restart apache2</code>	<code># Restart service</code>
<code>\$ sudo systemctl reload apache2</code>	<code># Reload conf</code>
<code>\$ sudo systemctl enable apache2</code>	<code># starts apache2 at boot time</code>
<code>\$ sudo systemctl disable apache2</code>	<code># stops apache2 at boot time</code>
<code>\$ sudo systemctl is-active apache2</code>	<code># Shows whether the service is active or not</code>
<code>\$ sudo systemctl is-enabled apache2</code>	<code># Shows whether the service is enabled or not</code>

15.COMPRESSION / ARCHIVES

```
$ tar cf home.tar home # Create tar named home.tar containing
home/ (11 tar examples)

$ tar xf # Extract the files
file.tar from
file.tar

$ tar czf file # Create a tar with
file.tar.gz le gzip
compression s

$ gzip file # Compress file and renames
to file.gz (untar gzip it
file)
```

16.PROCESS RELATED

```
$ # Display your currently
processes (many parameters to active
learn)

$ ps aux | grep # Find all process id related
'telnet' telnet to
process

$ pmap # Memory map of
(kernel,user memory process
etc)

$ top # Display all running
(30 processes
examples)

$ kill pid # Kill process with
pid id (types of mentioned
signals)

$ killall # Kill all processes named
proc

$ pkill # Send signal to a process
processname its with
name

$ # Resumes suspended jobs
bringing them to foreground (bg and fg)
command)

$ fg # Brings the most recent job
foreground to

$ fg # Brings job n to the
n foreground
```

17. SYSTEM

\$ uname -a	=> Display linux system information (refer uname command detail)
\$ uname -r	=> Display kernel version
\$ cat /etc/redhat-release	=> Show which version of redhat installed
\$ uptime	=> Show how long system running + load (learn uptime command)
\$ hostname	=> Show system host name
\$ hostname -i	=> Display the IP address of the host (all options hostname)
\$ last reboot	=> Show system reboot history (more examples last command)
\$ date	=> Show the current date and time (options of date command)
\$ cal	=> Show this month calendar (what more in cal)
\$ w	=> Display who is online (learn more about w command)
\$ whoami	=> Who you are logged in as (example + screenshots)
\$ finger user	=> Display information about user (many options of finger command)

18. HARDWARE

\$ dmesg	=> Detected hardware and messages (dmesg many more options)
\$ cat /proc/cpuinfo	=> CPU model
\$ cat /proc/meminfo	=> Hardware memory
\$ cat /proc/interrupts	=> Lists the number of interrupts per CPU per I/O device
\$ lshw	=> Displays information hardware configuration of the system
\$ lsblk	=> Displays block device information in Linux (sudo yum installed util-linux-ng)
\$ free -m	=> Used and free memory (-m MB) (free command in detail)

\$ lspci -tv	=> Show PCI devices (very useful)
to find vendor ids)	
\$ lsusb -tv	=> Show USB devices (read more)
lsusb	
options)	
\$ lshal	=> Show a list of all devices with
their	
properties	
\$ dmidecode	=> Show hardware info from the
BIOS (vendor details)	
\$ hdparm -i /dev/sda	# Show info about disk sda
\$ hdparm -tT /dev/sda	# Do a read speed test on disk sda
\$ badblocks -s /dev/sda	# Test for unreadable blocks on
disk	
sda	

19. STATISTICS

\$ top	=> Display and update the top cpu processes (30 example options)
\$ mpstat 1	=> Display processors related statistics (learn mpstat command)
\$ vmstat	=> Display virtual statistics (very useful performance tool)
\$ iostat 2	=> Display I/O statistics (2sec
Intervals) (more examples)	
\$ tail -n 500	=> Last 10
messages, (everyday use tail options)	kernel/syslog
\$ tcpdump -i	=> Capture all packets flows
interface eth1 (useful to sort network issue)	
\$ tcpdump -i eth0 'port 80'	=> Monitor all traffic on port
80 (HTTP)	
\$	=> List all open files
to all active processes. (sysadmin favorite command)	
\$ lsof -u	=> List files opened by
testuser user	specific
\$ free -m	=> Show amount of RAM
usage	(daily
command)	
\$ watch df -h	=> Watch changeable data
linux	continuously(interesting
command)	

